

RUI SOUSA

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EDUCATION

Northwestern University, Evanston, IL
PhD Candidate, Economics Department

2021 – 2027 (expected)

Nova School of Business and Economics, Lisbon, Portugal
MSc Economics, GPA 19/20 (A) - Novo Banco Master's Scholarship

2016 – 2019

RESEARCH ACTIVITY

The Aggregate Green Elasticity of Substitution (**Draft**)

- Develop a semi-structural framework to estimate the short-run aggregate elasticity of substitution between clean and polluting energy sources—a key parameter governing energy price pass-through and the costs of decarbonization
- Construct an equilibrium relationship between energy consumption and observable prices, and exploit cross-state variation in energy mixes to implement a shift-share IV strategy, estimating an aggregate elasticity of substitution of 0.5
- Derive and calibrate a mapping between sectoral and aggregate elasticities, finding that end users substitute toward electricity more readily than the electricity sector substitutes toward clean generation (0.80 versus 0.52)

Cleansing Oil (**Draft**)

- Estimate that a 10% increase in oil prices reduces CO₂-equivalent atmospheric concentrations by 0.10 ppm relative to trend
- Show that petroleum consumption declines across refined products, partially substituted toward low-quality coal and bioenergy
- Find that oil price shocks reduce solar investment through tighter financial conditions and increased uncertainty, while accelerating hybrid and EV adoption

Climate Damages Across the Globe (Ongoing)

- Estimate heterogeneous country-level responses to temperature shocks using a global panel of economic and climatic data
- Combine machine learning methods to identify the economic and geoclimatic characteristics driving the heterogeneous effects

PROFESSIONAL EXPERIENCE

Pacific Investment Management Company (PIMCO), Newport Beach, United States
Quantitative Research Analyst

Jun 2026 – Aug 2026

- Reviewed the literature studying AI and developed a HANK model to study its medium-run macroeconomic effects
- Developed an interpretable model to forecast aggregate S&P 500 earnings per share, reducing the mean squared error by 18% relative to a historical-mean benchmark and achieving 75% directional accuracy

London School of Economics, London, United Kingdom

Sep 2020 – Sep 2021

Research Assistant, Centre for Macroeconomics. Advisors: Profs. Benjamin Moll & Ricardo Reis

- Solved and simulated continuous-time heterogeneous-agent models numerically
- Modeled rare inflation events using data-driven threshold selection and MLE of light- and heavy-tailed distributions
- Analyzed microdata from consumer and professional surveys to study the formation and distribution of inflation expectations

European Central Bank, Frankfurt am Main, Germany

Jun 2018 – Sep 2020

Analyst, Crisis Management - Quantitative Cluster

- Contributed to a natural language early-warning system processing 1+ million daily observations from a global media database
- Analyzed liquidity provisions during stress events and the impact of banking policy on profitability
- Contributed to the impact assessment of Basel III on MREL requirements

Trainee, Stress-Test Modeling

- Applied regression discontinuity and difference-in-differences methods to supervisory banking panel data to estimate the effects of stress tests and capital requirements on bank risk-taking and profitability
- Contributed to the development of a credit risk assessment tool using historical loan portfolio data

TOOLS & SKILLS

- **Programming:** Python, R, MATLAB
- **Time Series:** State-Space Models, Nonlinear Models, Stochastic Volatility, Bayesian Inference, Forecasting
- **Econometrics:** Shift-share IV, DiD, RDD, Statistical Learning
- **Computing:** High-performance computing (SLURM), Git, $\LaTeX$

- **Quantitative Macro:** Heterogeneous-Agent Models, Continuous-Time Methods, Sequence-Space Jacobian, Dynare
- **Datasets:** Compustat, CRSP, CES, PSID, IBES, SPF, SCE, MER, SEDS, CMIP6
- **Languages:** English, Portuguese (fluent); French, Spanish (conversational)